Why Do Airline Ticket Prices Change So Dramatically?

An Economic Analysis of Dynamic Pricing and Consumer Welfare

Prepared for: Aviation Consumer Protection Policy Analysts, U.S. Department of Transportation
Secondary audience: General consumers interested in airline pricing
Report type: Informational Report
Date: October 30, 2026

I. Introduction

Anyone who has booked a flight knows how frustrating airline ticket prices can be. You find a flight at a reasonable price, decide to think about it for a few days, and return to discover that the same ticket now costs significantly more. Sometimes the opposite happens, and a flight becomes cheaper just days before departure. Even passengers sitting next to each other on the same plane may have paid hundreds of dollars apart for essentially the same trip. What causes these differences, and why do airline prices seem so unpredictable?

Although these fluctuations can feel arbitrary, they reflect a complicated set of economic decisions. Airlines must decide how much to charge today without knowing exactly how many passengers will book tomorrow, what competing airlines will offer, or how much future customers will be willing to pay. These uncertainties make airline pricing fundamentally different from simply setting a price based on the cost of providing a service.

For consumers, these price fluctuations have consequences beyond the frustration of not knowing when to book. The way airlines set fares can influence who can afford to fly, how travelers make purchasing decisions, and how the benefits of air travel are distributed between airlines and passengers. Understanding these effects is important not only for travelers but also for consumer protection policymakers responsible for evaluating airline pricing practices. This report examines how supply and demand, price discrimination, and airline competition influence ticket prices before considering their consequences for consumer welfare and pricing transparency.

II. Supply and Demand

The supply of seats on a particular flight is largely fixed once departure approaches. A Tuesday morning flight from Baltimore to Chicago in February may attract relatively little demand, while the same route on the Sunday after Thanksgiving could sell out weeks in advance. The cost of transporting an additional passenger may not differ dramatically between these flights, but the value of the available seats does. When demand is high relative to capacity, airlines have little

incentive to offer discounts. When demand is weak, lower fares can attract additional passengers and reduce the risk of departing with empty seats.

This constraint depends on the time horizon. An airline cannot easily increase capacity on a flight departing tomorrow, but sustained growth in demand may eventually justify larger aircraft, additional departures, or new routes. In the short run, however, prices must do much of the work of adjusting to unexpected demand. A sudden increase in bookings can raise fares even when the airline's operating costs have not changed.

The timing of purchases introduces another complication. Selling a seat today guarantees revenue, but it also prevents the airline from selling that seat to someone else tomorrow. Economists refer to the value of the next-best alternative as *opportunity cost*. For an airline, the opportunity cost of accepting a discounted booking depends on the likelihood that a higher-paying passenger will arrive before departure. This creates a tradeoff: selling too cheaply may sacrifice future revenue, while holding out for a higher fare risks leaving the seat empty.

Airlines therefore respond not only to current bookings but also to forecasts of future demand. Consider two flights with 20 seats remaining. If one is expected to attract only 10 additional customers, offering lower fares may help fill the aircraft. If the other is expected to attract 30 customers, the airline has a stronger incentive to preserve its remaining capacity. Even with the same number of empty seats, two flights can have very different ticket prices. These forecasts are uncertain, and unexpected changes in bookings can lead airlines to revise their prices as departure approaches.

The value of preserving a seat also depends on who is likely to purchase it. Some passengers can change their travel dates or choose another airline when fares rise, while others have little flexibility. These differences in willingness to pay help explain why airlines do not simply charge every passenger the same price.

III. Price Discrimination

A vacationer planning a trip several months in advance might be willing to change their travel dates to save $100. Someone flying to an important business meeting the next morning probably has much less flexibility. Although both passengers are purchasing transportation on the same aircraft, they place different values on that service. In economic terms, the vacationer's demand is generally more *price elastic*, meaning they are more likely to change their purchasing decision in response to a change in price.

Airlines take advantage of differences in price elasticity through *price discrimination*, charging different prices for comparable services based on differences in customers' willingness to pay rather than differences in production costs alone. Because airlines cannot directly observe how much each customer is willing to pay, they use purchasing behavior and ticket restrictions to

distinguish between types of travelers. Advance-purchase requirements, refundability, and restrictions on changing flights allow airlines to offer lower fares to price-sensitive passengers while charging more to those who need greater flexibility.

The resulting fare differences can be substantial. Borenstein and Rose (1994) found that the expected absolute difference between fares paid by two randomly selected passengers on the same airline and route was equivalent to 36% of the route's average fare. Surprisingly, they also found greater price dispersion on routes with more competition. Consider a route served by two competing airlines, Airline A and Airline B. A leisure traveler booking months in advance may be willing to fly with either carrier, encouraging both airlines to offer lower fares to attract their business. A business traveler booking just days before departure, however, may need a specific departure time that only Airline A offers. Airline A then can charge a substantially higher fare because the traveler has fewer practical alternatives. Competition can therefore lower prices for flexible passengers while leaving fares relatively high for those with less flexibility, widening the difference between the prices passengers pay. Although Borenstein and Rose's analysis uses data from 1986, their findings demonstrate how competition and price discrimination can operate together rather than simply offsetting one another.

Greater price dispersion does not, however, necessarily indicate that consumers are worse off. In the example above, competition could increase the variation in ticket prices simply by lowering fares for leisure travelers, without raising prices for business travelers. The average fare might actually decrease even as price dispersion increases. Lower fares could also make air travel affordable for passengers who otherwise would not have purchased a ticket, potentially increasing total consumer welfare. On the other hand, airlines may capture more of the benefits of air travel by charging higher fares to passengers with fewer alternatives.

Price discrimination can also occur over much shorter periods. Escobari, Rupp, and Meskey (2019) examined changes in posted airfares throughout the day and found evidence of higher prices during office hours, when business travelers may be more likely to book, and lower prices in the evening, when leisure travelers may be more active. By accounting for factors such as remaining inventory and time until departure, the researchers identified a component of these price differences consistent with differences in customers' willingness to pay rather than changes in seat availability alone. Although the estimated effect was relatively small, with office-hour fares averaging approximately $1 more seven days before departure, the finding suggests that the types of passengers shopping at different times may influence fares independently of changes in seat availability. The researchers also found that this component of price discrimination became more pronounced as departure approached, when the proportion of business travelers was expected to increase.

IV. Airline Competition and Dynamic Pricing

An airline's ability to charge different prices depends partly on the alternatives available to its passengers. Competition is therefore more complicated than simply counting how many carriers serve a route. Two airlines may both offer flights between Baltimore and Chicago, but a nonstop morning departure is not necessarily a close substitute for an evening flight with a connection. Differences in schedules, airport locations, and travel times can give an airline considerable pricing power over particular departures, even when several carriers operate between the same cities. Consequently, the presence of competing airlines does not guarantee that every passenger has a practical alternative.

Interestingly, airlines do not always wait for a competitor to enter a market before lowering their prices. Goolsbee and Syverson (2008) examined how established airlines responded when Southwest Airlines began serving both airports at the endpoints of a route but had not yet introduced flights between them. Because Southwest was already operating at both airports, the likelihood of it eventually connecting them increased, creating a credible threat to the airlines currently serving that route. The researchers found that existing airlines reduced fares significantly before Southwest began operating, with more than half of Southwest's total effect on existing fares occurring before its first flight on the route. These price reductions were concentrated on the routes Southwest was expected to enter, rather than extending to other routes served by the same airlines from nearby airports. Although the researchers could not conclusively determine whether the reductions were intended to discourage Southwest from entering or prepare for its arrival, the finding shows that airlines may face competitive pressure even before passengers have another carrier to choose from.

However, competition does not always lead airlines to make the most economically efficient pricing decisions. Betancourt et al. (2022) identified a counterintuitive problem called the *Bertrand scarcity trap*, in which competition can encourage airlines to sell seats too quickly. Consider two airlines competing for passengers several weeks before departure. Each has an incentive to offer lower fares to attract customers away from the other, even if doing so means selling seats that could be worth considerably more closer to departure. If both airlines aggressively compete for these early bookings, they may exhaust their capacity before later passengers have an opportunity to purchase tickets. This creates a problem because some of those later passengers may value the flight more highly than those who purchased discounted tickets earlier. Although lowering prices helps each airline compete for customers in the present, the resulting pattern of sales may prevent seats from reaching the passengers who value them most.

Using daily fare and booking data from competing U.S. airlines, Betancourt and colleagues estimated a model of dynamic price competition and compared its predicted equilibrium with alternative pricing rules based on practices observed at a major airline. These alternative rules generated approximately 4–5% higher airline revenue and 3% higher consumer surplus in their model. The results suggest that preserving more seats for later passengers can sometimes benefit

both airlines and consumers, even if fewer discounted tickets are available early in the booking period. This does not mean that higher fares or less competition generally improve consumer welfare; the estimated gains are modeled outcomes rather than observed industry-wide improvements.

V. Consumer Welfare and Pricing Transparency

Although dynamic pricing allows airlines to increase revenue by charging different fares, its effects on consumers are less straightforward. Economists often evaluate these effects using *consumer surplus*, the difference between what a passenger is willing to pay and the price they actually pay. Consider a business traveler willing to spend $600 on a flight. If the ticket costs $400, the traveler receives $200 in consumer surplus. If the airline instead charges $550, that surplus falls to $50, even though the passenger purchases the same flight. However, higher fares may also encourage airlines to preserve seats for passengers who need to book later, while lower advance-purchase fares can make travel affordable for more price-sensitive customers. Evaluating dynamic pricing therefore involves considering both the prices passengers pay and whether seats are available when they want to travel.

Williams (2022) investigated this tradeoff using flight-level airline data and an economic model comparing dynamic pricing with a system in which fares remain uniform over the booking period. His findings suggest that dynamic pricing generally benefits early-booking leisure travelers while reducing the consumer surplus of late-booking business travelers. However, the higher fares paid by business travelers also help preserve seat availability closer to departure, when those passengers may have fewer alternatives. Williams found that dynamic pricing increased aggregate welfare across the markets in his analysis, although the effects varied depending on the source of price changes. Because aggregate welfare includes benefits to both airlines and passengers, this improvement does not imply that all consumers benefit equally. In particular, the study found that late-booking business travelers generally received less consumer surplus under dynamic pricing than under uniform pricing.

Beyond the economic effects of changing fares, pricing transparency presents a separate consumer protection concern. Even when fare differences reflect changes in demand or pricing strategies, passengers need accurate information to compare their options. For example, a $180 ticket may initially appear cheaper than a $220 alternative, but baggage charges or fees associated with changing a reservation could reverse that comparison. The U.S. Department of Transportation requires advertised airfares to include mandatory taxes and fees, helping consumers identify the actual price of purchasing a ticket (U.S. Department of Transportation, 2025). However, understanding the total cost of travel also requires clear information about optional services and ticket restrictions. Greater transparency cannot eliminate the uncertainty of future fare changes, but it can help passengers make informed comparisons between the options currently available to them.

VI. Conclusion

Airline ticket prices are not simply determined by how full a plane is or how close it is to departure. They also reflect airlines' expectations of future demand, differences in passengers' willingness to pay, and the availability of competing flights. This explains why two passengers can pay dramatically different prices for the same trip, and why fares sometimes increase even when seats remain available.

For consumers, the clearest advantage is flexibility. Travelers who can compare multiple airlines, departure times, and travel dates are better positioned to take advantage of lower fares, while those with fixed schedules have fewer opportunities to avoid price increases. Booking early can help secure affordable options, particularly on routes with limited competition, but it does not guarantee the lowest price. Waiting for a discount means accepting the risk that cheaper fares or suitable flights will disappear. Consumers should also compare the total cost of tickets, including baggage fees and restrictions, rather than relying on the advertised base fare alone. Ultimately, understanding airline pricing cannot eliminate uncertainty, but it can help travelers recognize when flexibility, advance planning, and comparison shopping are most valuable.

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